Redox potential in human semen: Validation and qualification of the MiOXsys assay.

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Seminal oxidative stress (OS) is a major contributing factor to male infertility. Semen analysis cannot identify reactive oxygen species (ROS), which can be measured using a chemiluminescence assay. Measurement of redox potential provides a more comprehensive assessment of OS, although the test has yet to be fully validated. This study aimed to validate the MiOXsys analyser for measuring static oxidation-reduction potential (sORP). Results demonstrated that duplicate measurements must be taken, sensors must be batch tested, and sockets should be regularly changed to avoid inconsistency in measurement. Measurement of sORP using MiOXsys exhibited good reproducibility across different operators ($p = 0.469$), analysers ($p = 0.963$) and days ($p = 0.942$). It is not affected by mechanical agitation ($p = 0.522$) or snap freezing and thawing ($p = 0.823$). The stability of sORP over time requires further verification, particularly in samples with high initial sORP. Measurement is temperature sensitive between 2 and 37°C, significantly increasing with increasing temperature ($p = 0.0004$). MiOXsys is a more stable assay for assessing OS than chemiluminescence methods and permits greater flexibility for sample handling. MiOXsys could be implemented to complement semen analysis as part of routine diagnostic testing for male infertility and may be useful in identifying contributing factors to idiopathic infertility.

Seminal oxidation-reduction potential as a possible indicator of impaired sperm parameters in Turkish population.

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Oxidative stress (OS) has been shown to have a key role in male infertility. Recently, a new measurement method has been developed to measure the overall oxidation-reduction potential (ORP) in a semen sample known as the MiOXSYS system. The aim of this study was to investigate the correlation of sperm parameters with oxidative stress levels determined by ORP and to evaluate whether the current limit is able to distinguish abnormal sperm parameters from normal ones in Turkish population. Semen samples of 121 patients who applied for infertility investigation were divided into two groups as (OS +; n:39) and (OS -; n:82). Semen parameters were compared between groups. Sperm concentration, total motility and progressive motility were found to be significantly lower in OS (+) patients compared to those OS (-), while immotile sperm count was significantly higher in OS (+) patients. Oxidative stress determined by MiOXSYS system was found to be related to reduced sperm parameters in Turkish population, which may be used as an indicator of poor sperm parameters and a support to routine semen analysis. In addition, recommended reference value was found to be reliable in distinguishing normal from impaired semen parameters.

Oxidation-reduction potential of semen: what is its role in the treatment of male infertility?

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The diagnosis of male infertility relies largely on conventional semen analysis, and its interpretation has a profound influence on subsequent management of patients. Despite poor correlation between conventional semen parameters and male fertility potential, inclusion of advanced semen quality tests to routine male infertility workup algorithms has not been widely accepted. Oxidative stress is one of the major mediators in various etiologies of male infertility; it has deleterious effects on spermatozoa, including DNA damage. Alleviation of oxidative stress constitutes a potential treatment strategy for male infertility. Measurement of seminal oxidative stress is of crucial role in the identification and monitoring of patients who may benefit from treatments. Various tests including reactive oxygen species (ROS) assay, total antioxidant capacity (TAC) assay or malondialdehyde (MDA) assay used by different laboratories have their own drawbacks. Oxidation-reduction potential (ORP) is a measure of overall balance between oxidants and antioxidants, providing a comprehensive measure of oxidative stress. The MiOXSYS™ System is a novel technology based on a galvanostatic measure of electrons; it presents static ORP (sORP) measures with static referring to the passive or current state of activity between oxidants and antioxidants. Preliminary studies have correlated sORP to poor semen qualities. It is potentially useful in prognostication of assisted reproductive techniques outcomes, screening of antioxidants either in vivo or during IVF cycles, identification of infertile men who may benefit from treatment of oxidative stress, and monitoring of treatment success. The simplified laboratory test requiring a small amount of semen would facilitate clinical application and research in the field. In this paper, we discuss the measurement of ORP by the MiOXSYS System as a real-time assessment of seminal oxidative stress, and argue that it is a potential valuable clinical test that should be incorporated into the male infertility workup and become an important guide to the treatment of oxidative stress-induced male infertility.

MiOXSYS: a novel method of measuring oxidation reduction potential in semen and seminal plasma.

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To measure oxidative reduction potential (ORP) in semen and seminal plasma and to establish their reference levels. ORP levels were measured in semen and seminal plasma. Tertiary hospital. Twenty-six controls and 33 infertile men. None. Static ORP (sORP) and capacitance ORP (cORP) were measured in semen and seminal plasma at time 0 and 120 minutes. Correlation of ORP was assessed between [1] semen and seminal plasma and [2] time 0 and 120 minutes. The association with sperm parameters was studied in (a) controls and (b) infertile patients, and a receiver operating characteristic curve was generated to establish the sORP cutoff. Semen sORP and cORP levels were associated with seminal plasma levels at time 0 and time 120 minutes. In controls and infertile patients, an inverse relationship of sORP levels was established with concentration and total sperm count in semen as well as seminal plasma at time 0 and 120 minutes. Classification of subjects based on sperm motility showed that subjects with abnormal motility present with poor concentration, total count, morphology, and elevated levels of semen and seminal plasma sORP at time 120 minutes. The sORP cutoff of 1.48 in semen and 2.09 in seminal plasma based on motility was able to distinguish subjects with normal semen quality from those with abnormal semen quality. The MiOXSYS System can reliably measure ORP levels in semen and seminal plasma. ORP levels are not affected by semen age, making this new technology easy to employ in a clinical setting.

Cumene hydroperoxide induced changes in oxidation-reduction potential in fresh and

frozen seminal ejaculates.

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Oxidation-reduction potential (ORP) is a newer integrated measure of the balance between total oxidants (reactive oxygen species-ROS) and reductants (antioxidants) that reflects oxidative stress in a biological system. This study measures ORP and evaluates the effect of exogenous induction of oxidative stress by cumene hydroperoxide (CH) on ORP in fresh and frozen semen using the MiOXSYS Analyzer. Semen samples from healthy donors (n = 20) were collected and evaluated for sperm parameters. All samples were then flash-frozen at -80°C. Oxidative stress was induced by CH (5 and 50 µmoles/ml). Static ORP (sORP-(mV/106 sperm/ml) and capacity ORP (cORP-µC/106 sperm/ml) were measured in all samples before and after freezing. All values are reported as mean ± SEM. Both 5 and 50 µmoles/ml of CH resulted in a significant decline in per cent motility compared to control in pre-freeze semen samples. The increase in both pre-freeze and post-thaw semen samples for sORP was higher in the controls than with 50 µmoles/ml of CH. The change from pre-freeze to post-thaw cORP was comparable. The system is a simple, sensitive and portable tool to measure the seminal ORP and its dynamic impact on sperm parameters in both fresh and frozen semen specimens.

Relationship of Seminal Oxidation-Reduction Potential with Sperm DNA Integrity and pH in Idiopathic Infertile Patients.

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Seminal oxidative stress (OS) is one of the most promising factors to describe the causes of idiopathic male infertility. Redox balance is essential in several biological processes related to fertility, so alterations such as high reactive oxygen species (ROS) levels or low antioxidant agent levels can compromise it. MiOXSYS has been developed to evaluate the seminal static oxidation-reduction potential (sORP) and it has been proposed as an effective diagnostic biomarker. However, its relationship with parameters like sperm DNA fragmentation (SDF), chromatin compaction status or seminal pH requires further analysis, making it the object of this study. Semen and sORP analysis were performed for all samples. A terminal deoxynucleotidyl transferase dUTP nick end labeling assay (TUNEL) and Comet assay were used to assess SDF and chromomycin a3 (CMA3) test to assess sperm chromatin compaction. Regarding sORP measures, it was found that alkaline pH has an effect on sample reproducibility. To our knowledge, this unexpected effect has not been previously described. A statistical analysis showed that sORP correlated negatively with CMA3 positive cells and sperm motility, but not with SDF. As redox dysregulation, which occurs mainly at the testicular and epididymal level, causes chromatin compaction problems and leaves DNA exposed to damage, an excess of ROS could be counterbalanced further by a seminal supply of antioxidant molecules, explaining the negative correlation with CMA3 positive cells but no correlation with SDF. Our results show that the study of idiopathic infertility would benefit from a combined approach comprising OS analysis, SDF and chromatin compaction analysis.

Reassessing the interpretation of oxidation-reduction potential in male infertility.

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Male Infertility Oxidative System (MiOXSYS) has been proposed as a rapid and promising technology for the evaluation of sperm oxidative stress. In this case-control study, 134 men with normal sperm parameters (NSP) and 574 men with abnormal sperm parameters (ASP), according to the World Health Organization sperm assessment references values established in 2010, were enrolled. Conventional sperm parameters were evaluated in all patients. Sperm static oxido-reduction potential (sORP) was assessed using the MiOXSYS. Sperm DNA integrity was measured in 604 patients. To ensure that sperm concentration was not a confounding factor in the sORP index ratio, sperm and seminal fluid sORP from 57 randomly selected additional patients were also measured using the MiOXSYS. sORP index (mV/106 sperm/mL) was higher in patients with ASP and seemed to correlate with conventional sperm parameters. Although receiver-operating characteristic analysis revealed that a sORP index cut-off of 0.79 could differentiate normal from ASP with 57.7% sensitivity and 73.1% specificity, these values are much lower than those found in the literature. These values also need to be higher to be applicable in a clinical setting. Furthermore, absolute sORP (mV) was not different in the presence or absence of spermatozoa. sORP index relationships with sperm parameters seem rather be due to sperm concentration, denominator of the sORP index ratio. The establishment of a reliable method using the absolute sORP value, independent of sperm concentration, needs to be addressed. Other oxidative stress biomarkers could be used to validate this method. The World Health Organization (WHO) has recognized that oxidative stress may have a role in male infertility. Oxidative stress happens when there is an imbalance between the production of molecules containing oxygen and the antioxidants, molecules that neutralize the molecules containing oxygen. The molecules containing oxygen can cause damage to sperm DNA. This damage can be measured using a particular index and this study looked at whether the concentration of the sperm sample might have an impact on results and suggests this should be taken into consideration by clinicians and researchers.

Aging increases oxidative stress in semen.

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As age increases, oxidative stress increases, sperm motility decreases, and DNA fragmentation increases. To date, reports of age-related effects on semen have focused on reactive oxygen species (ROS) or total antioxidant capacity (TAC) as indicators of oxidative stress. However, assessments of ROS and TAC must be considered within a more comprehensive context in order to correctly evaluate oxidative stress and interpret findings. In this regard, the purpose of this study was to investigate the relationship between the static oxidation reduction potential (sORP) and paternal age with the goal of using the sORP as an indicator of semen oxidative stress. Semen samples from 173 men were analyzed for the following parameters: volume, motility, and beat cross frequency (BCF). The sORP was measured by using the MiOXSYS™ system. The correlation between semen parameters and the sORP level was analyzed as a function of age. The rate of sORP positivity was compared between men <34 and ≥ 34 years of age, with a positive sORP defined as a level ≥ 1.38. Volume, motility, and BCF were negatively correlated with age ($p<0.001$). The semen sORP level was positively correlated with age ($p<0.05$). The rate of sORP positivity was significantly increased in men ≥ 34 years of age compared with that in men <34 years of age (33% compared with 12%, respectively; $p<0.01$). The sORP may play a role in age-related decreases in semen parameters (volume, motility, and BCF). The rate of sORP positivity increased significantly after 34 years of age.

Relation between semen oxidative reduction potential in initial semen examination and

IVF outcomes.

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The MiOXSYS system is a new technique to analyze the semen oxidative reduction potential (ORP) that may use to classify the level of sperm DNA integrity. It does not clearly explain how the semen ORP values could help to change the IVF outcomes. We have analyzed correlations between semen ORP value and the IVF results. Four hundred and thirty couples were enrolled. The male counterparts were divided into two groups according to their semen ORP values and compared the fertilization rate, cell cleavage rate, and embryo quality, following the IVF procedures. The relations between ORP values and the clinical pregnancy, live birth, and abortion rates were analyzed. The ORP values show negative and positive correlations with some conventional semen parameters. The fertilization and the cleavage rate did not show any differences in those two groups, but the transferable embryo rate was significantly high in patients with high semen ORP. However, the patients with high ORP show a tendency to lower clinical pregnancy with a low abortion rate compared to the low ORP group. The main purpose of measuring the ORP value in semen is still questionable and shows controversial results.

Safety of Tenofovir Disoproxil Fumarate Among Breastfeeding Infants of Patients With Chronic Hepatitis B: A Systematic Review.

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An integral component to achieving worldwide chronic hepatitis B (CHB) elimination is addressing vertical transmission. Guidelines differ in their recommendations for breastfeeding while on tenofovir disoproxil fumarate (TDF). To conduct a systematic review of published studies analysing the concentration of tenofovir (TFV) in the breast milk of mothers receiving TDF and determining infant exposure from breastfeeding. We conducted a systematic literature search of studies evaluating infant safety from the breast milk of breastfeeding mothers receiving TDF for any indication that reported a TFV breast milk concentration. Daily infant exposure was used to calculate the relative dose of TFV in infants. Other pertinent information collected was the concentration of TFV in maternal and infant plasma, the duration of therapy of TDF and the indication for TDF. We identified 10 studies including 443 patients-266 of whom were mothers, and the remaining were infants-that reported the TFV concentration of breast milk in breastfeeding mothers receiving TDF. A total of 654 breast milk samples were included. The mean TFV concentration from all the studies that reported a median concentration of TFV was 4.8 ng/mL (95% CI [3.8, 5.8]). The mean infant exposure of TFV from breast milk was 0.56 μg/kg/day (95% CI [0.44, 0.68]). The mean relative dose was determined to be 0.01% of the weight-based recommended infant dose. Infant plasma levels of TFV were also collected. This was undetectable in a majority of the studies that reported it. Based on the negligible infant exposure of TFV while breastfeeding, from a pharmacologic and toxicity standpoint, maternal dosing of TDF appears safe for breastfeeding infants.